

Host-Based Ransomware Detection and Mitigation

1. Host-Based Ransomware Mitigation and Detection without Third-Party Tools

1.1 Detect and Kill Malicious Processes

- Use system commands to identify and terminate processes linked to ransomware.
- Commands:
 - Windows: `taskkill /IM [process_name] /F`
 - Linux: `kill -9 [PID]`
- Script Example (Linux): Detect high CPU processes and terminate if suspicious.

1.2 Monitor and Detect Suspicious File Modifications

- Use `inotifywait` (Linux) or PowerShell (Windows) to detect rapid or unauthorized file changes.

1.3 Restrict Permissions on Sensitive Directories

- Limit write permissions on critical directories.
- Commands:
 - Linux: `chmod -R 400 /home/user/important_folder`
 - Windows: `icacls "C:\sensitive_directory" /deny "Everyone:(W,D)"`

1.4 Use Built-in Event Logging and Monitoring

- Windows: Monitor Event ID 4688 for new process creation events.
- Linux: Use `syslog` to track unauthorized activities.

1.5 Configure Firewall Rules for Outbound Connections

- Block unusual outbound traffic from local malware processes.

- Commands:

- Windows Firewall: ``New-NetFirewallRule -DisplayName "Block Outbound" -Direction Outbound -Action Block``
- Linux (iptables): ``sudo iptables -A OUTPUT -d [IP] -j DROP``

2. Host-Specific Ransomware Mitigation Techniques

2.1 Process Whitelisting and Behavior Monitoring

- Only allow verified processes to run and terminate unknown executables.
- Windows: Use AppLocker or PowerShell for whitelist policies.

2.2 File Integrity Monitoring (FIM)

- Regularly monitor critical files to detect unauthorized modifications.
- Commands:
 - Windows: PowerShell hash monitoring.
 - Linux: Use md5sum with cron jobs.

2.3 Restrict Access to Critical Files and Directories

- Limit access to sensitive files using file permissions.

2.4 Controlled Folder Access (Windows-Specific)

- Enable Controlled Folder Access to protect critical directories.

2.5 Regular Backups with Versioning

- Perform regular backups to local storage with versioning to restore clean data.

2.6 System Restore Points (Windows)

- Schedule system restore points to roll back changes made by ransomware.

2.7 Monitor Unusual Disk Activity

- Ransomware spikes disk I/O due to file encryption. Use ``iostat`` (Linux) or PowerShell to monitor.

2.8 Disable Macros in Office Applications

- Block macros in Office apps to reduce the risk of infection.

2.9 Harden Local User Accounts

- Limit access to critical files and administrative privileges for users.